

CDFD Runtime Paper 11: Validation, Precision, and Falsifiability

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Abstract

Executive synthesis. CDFD is useful only if it can fail. This paper defines the validation standard for the runtime series: numerical sanity checks, finite-value audits, reproducible CLI commands, experiment selection, precision analysis, and domain-specific falsification. The local experiments are treated as model diagnostics unless external validation is available.

Greek-symbol convention. Unless a manuscript states a tighter equation-local meaning, Greek symbols are local CDFD variables or coefficients, not universal constants. Here Φ denotes driving flux, Ψ_s denotes the adaptive operating ratio, Ω denotes a domain or state space, and Λ denotes the Life Number where used. The coefficients α , β , and γ denote accumulation or reinforcement, relaxation or decay, and diffusion, coupling, or re-distribution. The symbols δ and ϵ denote perturbation or tolerance scales, while η , λ , μ , ν , ρ , σ , τ , and ϕ denote local efficiency, rate, baseline, frequency, density or correlation, noise or variance, time-scale, and phase or field terms. Subscripts restrict any coefficient to the named pathway, tissue, domain, or experiment.

Runtime evidence path

Prior CDFD mathematics $\longrightarrow$ CDFL/runtime state $\longrightarrow$ bounded output $\longrightarrow$ falsification target.

Figure 1: Conceptual schematic for the runtime papers. The engine translates the mathematical frame from the prior CDFD papers into executable CDFL/runtime diagnostics; candidate outputs remain tied to explicit tests before any stronger claim is made.

1 Prior CDFD Basis

This manuscript does not rederive the mathematical core of CDFD. It uses the Mujjabi Adaptive Operating Ratio and the constrained-vacuum notation introduced in Part I [1],

the torus-knot hierarchy and boundary-mode work from the Part I sequence [2, 3], the public balance law developed in the Part I capstone [4], the Part I transport tests [5], and the Life Number and tri-regime biology bridge from Part II [6]. The runtime papers therefore act as an execution layer: they keep the mathematics connected to commands, outputs, falsification tests, and domain-specific limits.

2 Validation Ladder

The runtime uses a staged ladder:

1. source compiles and imports;
2. unit tests pass;
3. CDFL model validates and reports zero blocking diagnostics;
4. CDFL lint, format, and AST tooling agree with the parser surface;
5. simulation outputs are finite;
6. rerun reproduces the qualitative regime;
7. parameter sensitivity is documented;
8. independent data or experiment challenges the claim.

3 CLI Verification

The CLI pass added a reproducible command surface. A minimal verification set is:

```
python cdfd.py info
python cdfd.py domains
python cdfd.py diagnostics
python cdfd.py demo origins_of_life \
    --source-scenario mixed_source_surface_trap
python cdfd.py cdfl lint examples/heat_flow.cdfl
python cdfd.py cdfl format examples/heat_flow.cdfl
python cdfd.py cdfl ast examples/heat_flow.cdfl --json
python cdfd.py validate examples/heat_flow.cdfl
python cdfd.py run examples/heat_flow.cdfl \
    --out outputs/heat_flow_run.json
```

The result envelope includes provenance and `finite_audit`. Non-finite values are blockers unless the run is explicitly a boundary-failure test.

4 Test Evidence

The focused runtime check recorded:

```
python -m pytest -q tests/test_cli_runtime.py \
    tests/test_release_surfaces.py \
    tests/test_cdf1_vscode_extension.py
```

with 38 passing focused tests. These tests cover CLI runner behavior, CDFL parsing and execution, CDFL diagnostics/formatting/AST/sample helpers, Part II diagnostics export, origins-of-life guardrails, provider-key redaction, release-surface metadata, and the VS Code extension package metadata. A separate Extension Development Host smoke exercised the `.cdf1` language mode, commands, diagnostics, completions, symbols, formatting, validation, and runtime execution.

5 Precision Analysis

`experiments/reports/PRECISION_ANALYSIS.md` argues that ordinary float64 hardware is not the limiting factor for these simulations. The dominant errors are time-step selection, grid resolution, model stiffness, and CODATA measurement uncertainty for $\chi^* = \alpha^{-1} = 137.035999177$. Therefore, validation focuses on numerical convergence and model controls rather than on the laptop used to run the experiment.

6 Experiment Selection

`experiments/reports/FINAL_RELEASE_EXPERIMENT_SELECTION.md` is the current release-facing filter. It selects:

- the Mujjabi Vacuum Hysteresis Test as a candidate memory protocol;
- the Mujjabi $n = 7$ Knot Stability Test as a weak self-stabilization stress test;
- the Mujjabi Life Number Boundary Test as a bridge to Part II;
- frontier and gigamarathon sweeps as discovery coverage evidence;
- `part_ii_runtime_diagnostics.json` as the CLI-exported source-mix and guardrail bundle.

It also excludes overloaded adaptive-surface output as support; that run is an instability boundary.

7 Domain Falsification

Each domain needs explicit failure conditions. For example:

- OOL: if the proposed Λ boundary is indistinguishable from an energy-only model, the tri-regime extension adds no value.
- Finance-climate: if a directed edge remains after coupling is removed, the causal result is an artifact.
- Biology/pharmacology: if the Ψ_s proxy does not stratify outcomes beyond standard variables, the mapping fails.
- Vacuum hysteresis: if residual signatures vanish after thermal and QED confounders are controlled, physical vacuum-memory interpretation fails.

8 Conclusion

Validation is not a final chapter. It is the discipline that must accompany every paper, every experiment, and every CLI output. The runtime becomes more useful by becoming more falsifiable.

References

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